New York. Rather, we believe his laws work in exactly the same way regardless of where you are. Even though you have changed location, the laws that govern your body's motion remain as unaffected as the appearance of a cue ball that has been rotated.

This symmetry is known as *translational symmetry* or *translational invariance*. It applies not only to Newton's laws but also to Maxwell's laws of electromagnetism, to Einstein's special and general relativities, to quantum mechanics, and to just about any proposal in modern physics that anyone has taken seriously.

Notice one important thing, though. The details of your observations and experiences can and sometimes will vary from place to place. Were you to perform your gymnastics routines on the moon, you'd find that the path your body took in response to the same upward jumping force of your legs would be very different. But we fully understand this particular difference and it is already integrated into the laws themselves. The moon is less massive than the earth, so it exerts less gravitational pull; as a result, your body travels along different trajectories. And this fact—that the gravitational pull of a body depends on its mass—is an *integral* part of Newton's law of gravity (as well as of Einstein's more refined general relativity). The difference between your earth and moon experiences doesn't imply that the law of gravity has changed from place to place. Instead, it merely reflects an environmental difference that the law of gravity already accommodates. So when we said that the known laws of physics apply equally well in Connecticut or New York—or, let's now add, on the moon—that was true, but bear in mind that you may need to specify environmental differences on which the laws depend. Nevertheless, and this is the key conclusion, the explanatory framework the laws provide is not at all changed by a change in location. A change in location does not require physicists to go back to the drawing board and come up with new laws.

The laws of physics didn't have to operate this way. We can imagine a universe in which physical laws are as variable as those of local and national governments; we can imagine a universe in which the laws of physics with which we are familiar tell us nothing about the laws of physics on the moon, in the Andromeda galaxy, in the Crab nebula, or on the other side of the universe. In fact, we don't know with absolute